



ELEMENTARY MATH (CALCULUS)



HOW TO GRAPH FUNCTIONS

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LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- Identify function types.
- Highlight function characteristics.
- Analyze function behavior.
- Solve equations and inequalities.





WHAT ARE FUNCTIONS?

Before we dive into how to graph the various types of functions, let's have a recap. Functions are fundamental in mathematics, serving as tools to model various phenomena and solve real-world problems. As the word implies, it serves as the role one has to play.





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QUESTION

What is the function of a bed?



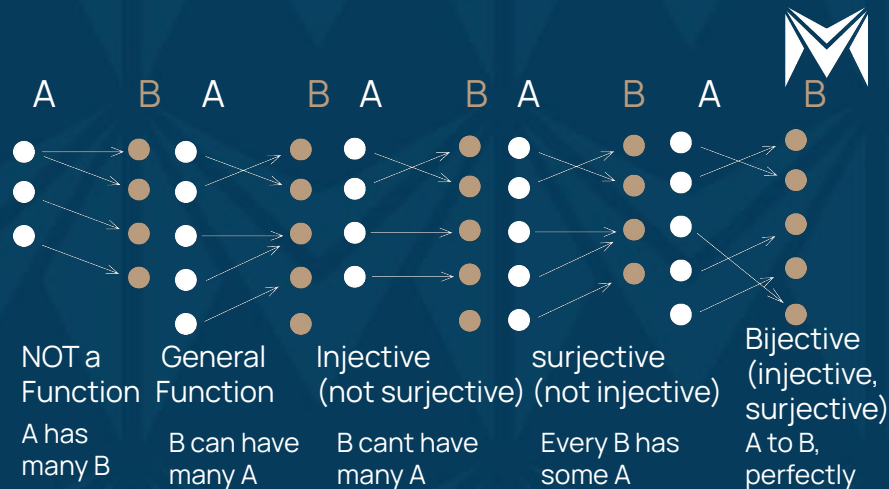


THE RELATIONSHIP BETWEEN SET AND FUNCTION

In Elementary Mathematics 1, you learnt about Set. A set is a collection of well-defined objects called members or elements of the set e.g. a collection of vowels, $\{a, e, i, o, u\}$ is a set. Sets help us define functions—the link between them. You also learnt about injective, surjective bijective, and composite functions (special types of functions).

Before we dive into the types of functions, let's learn a bit about the vertical line test.

$$(f \circ g)(x) = f(g(x))$$





VERTICAL LINE TEST

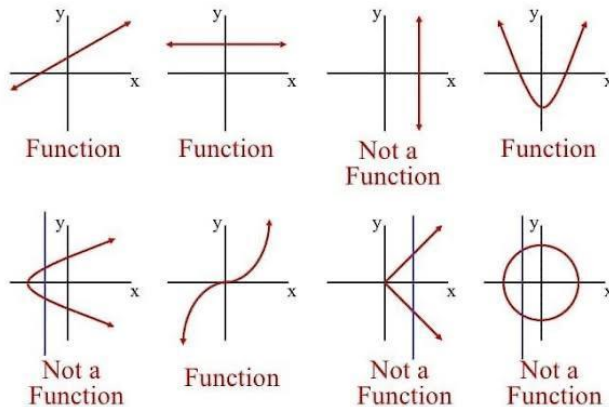
A vertical line test helps us to determine whether a curve on a graph is a function or not.

If any vertical line intersects the graph at more than one point, it fails the Vertical Line Test but if every vertical line intersects the graph at most once, it passes the Vertical Line Test.

Passing the Vertical Line Test indicates that the graph represents **a function**.

Failing the test means the graph does not represent a function (instead it represents **a relation**) because it associates one input with multiple outputs.

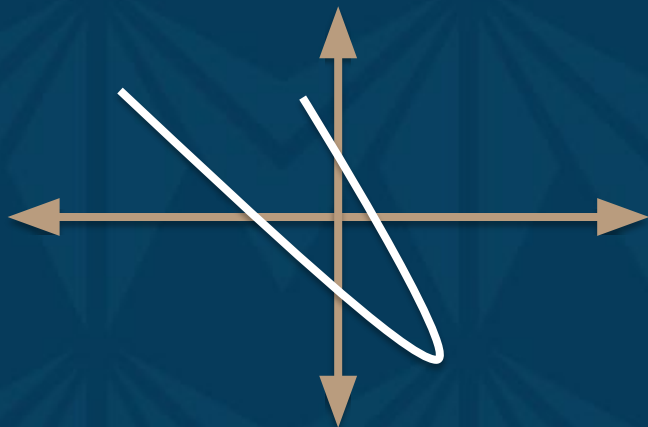
Vertical Line Test - Functions





TASK

Does this graph pass the vertical line test?

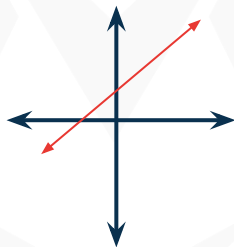




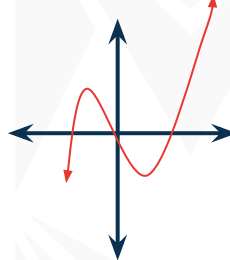
TYPES OF FUNCTIONS

- Linear Function
- Quadratic Function
- Exponential Function
- Logarithmic Function
- Absolute Value Function
- Piecewise Function
- Trigonometric Function
- Polynomial Function
- Rational Function
- Cubic Function etc.

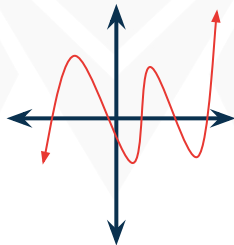
$$f(x) = mx + b$$



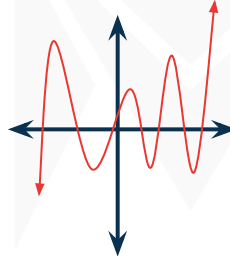
$$f(x) = ax^3 + \dots$$



$$f(x) = ax^5 + \dots$$



$$f(x) = ax^7 + \dots$$





LINEAR FUNCTION

Linear functions are a fundamental type of mathematical relation that represents a straight-line relationship between two variables.

- The general form of a linear function is $f(x) = mx + b$, where:
- m is the slope, determining the steepness of the line.
- b is the y-intercept, representing the value of $f(x)$ when $x = 0$.

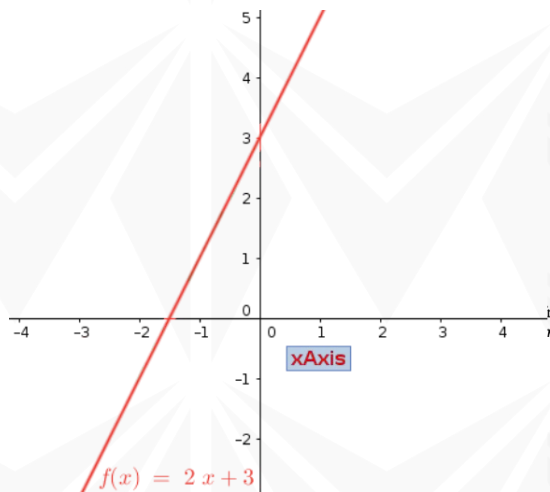


EXAMPLE OF A LINEAR FUNCTION

$$f(x) = 2x + 3$$

Recall from the equation of a straight line: $f(x) := mx + b$

Slope (m) = 2, Y-intercept (b) = 3





APPLICATIONS IN REAL LIFE:

Economics:

- Linear functions model concepts like cost, revenue, and profit in business and economics. Example: Cost (C) as a function of quantity (q): $C(q) = 2q + 1000$.

Physics:

- Linear functions describe simple physical relationships, such as distance (d) traveled over time (t).

Example:

- $d(t) = 50t + 100$ for constant velocity motion.

Engineering:

- Linear functions help engineers analyze structural elements, like stress (S) as a function of force (F).

Example:

- $S(F) = 2F + 500$ for linear material behavior



QUADRATIC FUNCTION

Quadratic functions are a type of mathematical relation that produce a parabolic curve when graphed.

The general form of a quadratic function is $f(x) = ax^2 + bx + c$, where:

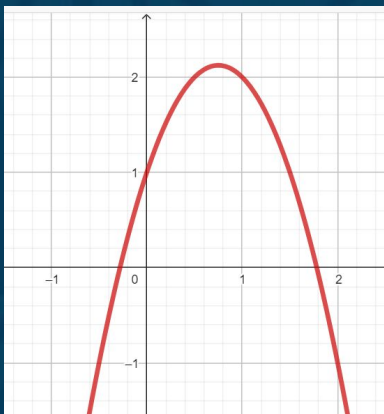
- a determines the direction and steepness of the parabola.
- b shifts the parabola horizontally.
- c is the y -coordinate of the vertex, which affects its vertical position.



EXAMPLE OF A QUADRATIC FUNCTION

$$g(x) = -2x^2 + 3x + 1$$

- -2 determines the direction and steepness of the parabola.
- 3 shifts the parabola horizontally.
- 1 is the y-coordinate of the vertex, which affects its vertical position





REAL-WORLD APPLICATIONS:

- **Engineering:** Engineers use quadratic functions to design curved structures, such as bridges and arches.

Example: $y(x) = 0.005x^2 - 0.1x + 5$ for the shape of an arch.

- **Economics:** Quadratic functions can represent cost functions, helping businesses optimize production.

Example: $c(x) = 0.02x^2 - 5x + 2000$ for manufacturing cost (C).



EXPONENTIAL FUNCTION

Exponential functions are mathematical functions in which a fixed number (the base) is raised to a variable exponent.

The general form of an exponential function is $f(x) = b^x$, where:

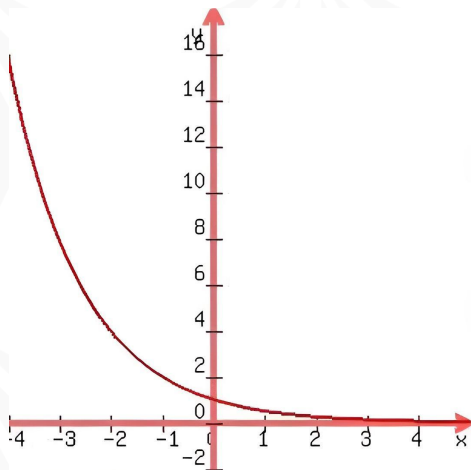
- x is the independent variable.
- b is the base, which determines the rate of growth or decay.



EXAMPLE OF AN EXPONENTIAL FUNCTION

$$g(x) = 0.5^x$$

- 0.5 is the base, which determines the rate of growth or decay





USE CASES IN GROWTH AND DECAY:

Population Growth: Exponential functions model population growth when resources are abundant.

- Example: $P(t) = P_o \cdot e^{rt}$ for population (P) with a fixed growth rate (r).

Compound Interest: Exponential functions calculate compound interest over time.

- Example: $A(t) = P \cdot (1 + nr)^{nt}$ for the final amount (A) with interest rate (r) compounded n times per year.

Radioactive Decay: Exponential functions describe the decay of radioactive materials.

- Example: $N(t) = N_o \cdot e^{-\lambda t}$ for the remaining quantity (N) of a substance.

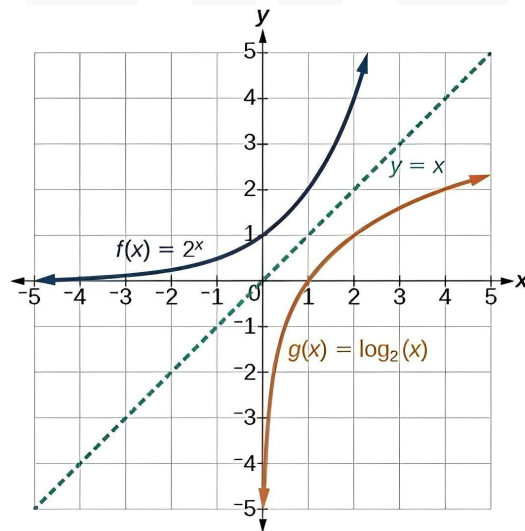


LOGARITHMIC FUNCTION

Logarithmic functions are mathematical functions that deal with the inverse operations of exponentiation.

The general form of a logarithmic function is $f(x) = \log_b(x)$, where:

- b is the base, defining the logarithmic scale.
- x is the argument, representing the number for which the logarithm is calculated.





EXAMPLES OF LOG FUNCTIONS:

Example 1:

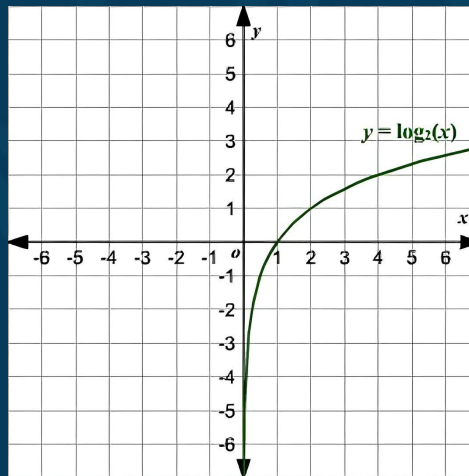
$$f(x) = \log_{10}(x)$$

– Common logarithm

Example 2:

$$g(x) = \log_e(x)$$

– Natural logarithm





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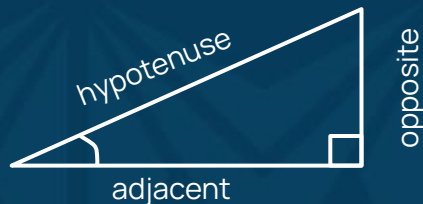


TRIGONOMETRIC FUNCTIONS



TRIGONOMETRIC FUNCTIONS

Sine (*sin*), cosine (*cos*), tangent (*tan*), Cosecant (*csc*), Secant (*sec*) and Cotangent (*cot*) are fundamental trigonometric functions used to describe relationships in triangles and periodic phenomena.



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\csc \theta = \frac{\text{hypotenuse}}{\text{opposite}}$$

$$\sec \theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

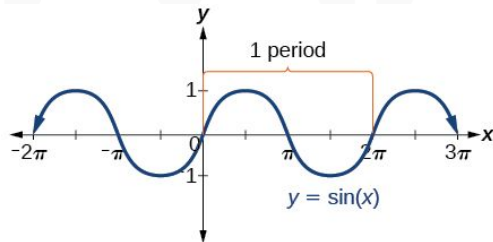
$$\cot \theta = \frac{\text{adjacent}}{\text{opposite}}$$



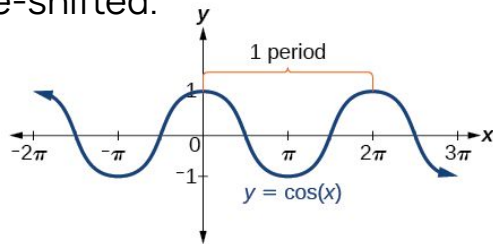


GRAPHS AND PERIODICITY:

- **Sine Function (*sin*):** Periodic with a period of 2π . Graph oscillates between -1 and 1.



- **Cosine Function (*cos*):** Also periodic with a period of 2π . Graph resembles the sine function but is phase-shifted.





APPLICATIONS IN GEOMETRY AND PHYSICS:

- **Geometry:** Trigonometric functions help measure angles and calculate side lengths in triangles. Used in trigonometry and geometry for surveying, navigation, and more.
- **Physics:** Trigonometric functions model periodic motion, such as the oscillation of a pendulum. Essential in physics to describe waveforms, sound, and light.





PIECEWISE FUNCTIONS

Piecewise functions are mathematical functions that consist of multiple "pieces" or segments, each defined by a different formula or expression. They are used to represent different relationships or rules for different intervals of the input variable.

Structure:

A piecewise function is typically defined using braces or conditional statements. For example:

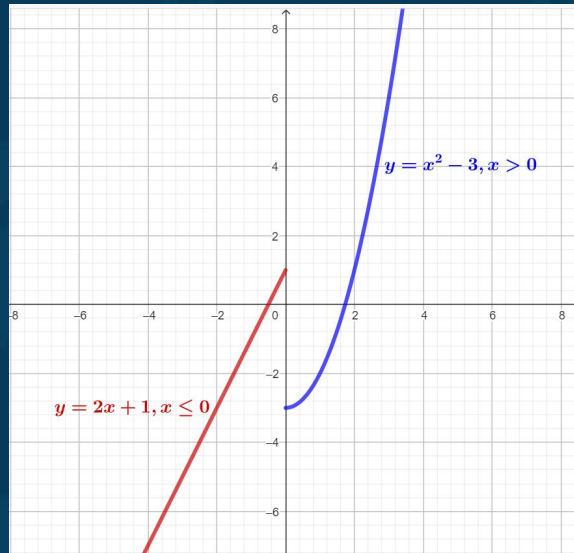
$$f(x) = \begin{cases} 2x + 1 & \text{if } x \leq 0 \\ x^2 - 3 & \text{if } x > 0 \end{cases}$$



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EXAMPLES AND GRAPHS:

$$f(x) = \begin{cases} 2x+1 & \text{if } x \leq 0 \\ x^2-3 & \text{if } x > 0 \end{cases}$$





VERSATILITY IN MODELING COMPLEX SCENARIOS:

Piecewise functions are versatile and can accurately model complex real-world scenarios with changing rules.

Examples: Modeling temperature variation throughout the day, pricing structures, and more.



ABSOLUTE VALUE FUNCTION

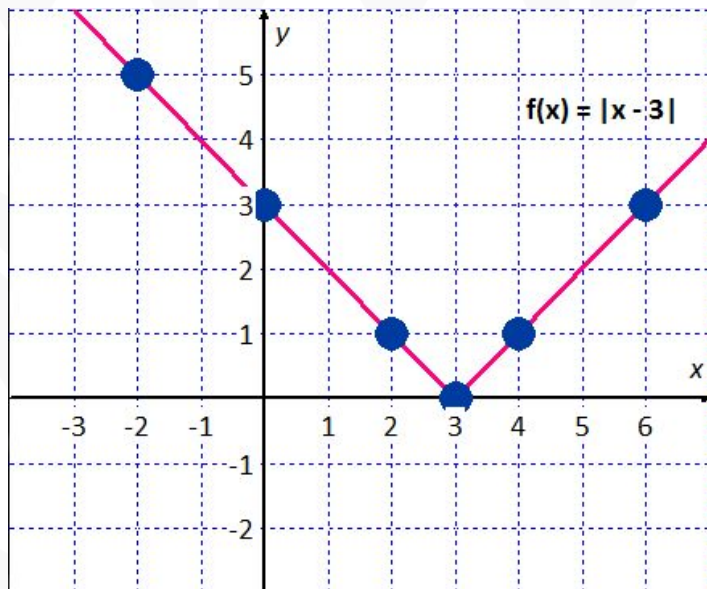
Absolute value functions are mathematical functions that determine the distance or magnitude of a number from zero on the number line. The absolute value of a number x , denoted as $|x|$, is defined as:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



EXAMPLE AND GRAPH:

$$f(x) = |x - 3|$$





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POLYNOMIAL FUNCTIONS



POLYNOMIAL FUNCTIONS

The general form of a polynomial function is

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0, \text{ where:}$$

- n is a non-negative integer (degree).
- $a_n, a_{n-1}, \dots, a_2, a_1, a_0$ are (coefficients).

Degrees and Leading Coefficients:

constants

Degree (n): The highest power of the variable x in the polynomial.

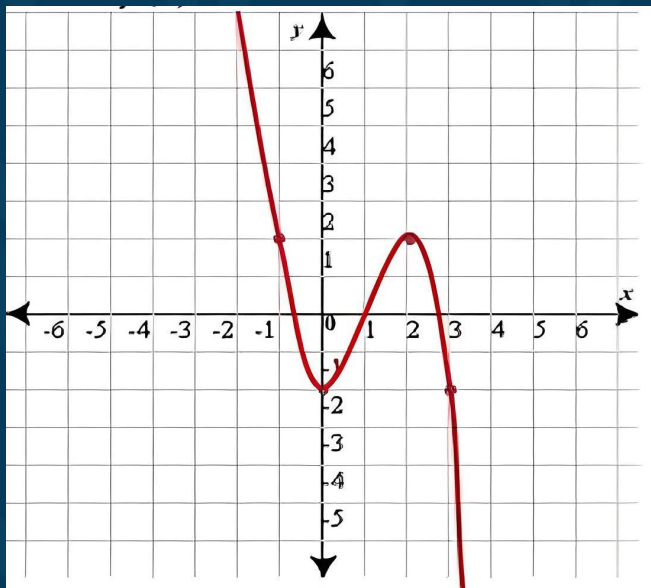
Leading Coefficient (a_n): The coefficient of the term with the highest power of x .



EXAMPLE AND GRAPH:

$$f(x) = -x^3 + 3x^2 - 2$$

Degree (n) = 3, Leading Coefficient (a_n) = -1





USE CASES IN MATHEMATICS AND ENGINEERING:

- **Mathematics:**

Polynomial functions are fundamental in calculus, algebra, and numerical analysis. They are used to approximate more complex functions and study their properties.

- **Engineering:**

Engineers use polynomial functions to model and analyze various systems and physical phenomena.

Examples: Control systems, signal processing, and structural analysis.



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SUMMARY



SUMMARY

- Linear functions: produce a straight line when graphed and are described by equations in the form $f(x) = mx + c$
- Quadratic functions: create parabolic shapes and are defined by equations like $f(x) = ax^2 + bx + c$
- Exponential functions: exhibit rapid growth or decay and have equations such as $f(x) = a \cdot b^x$
- Logarithmic function: are the inverse of exponential functions and are denoted $f(x) = \log_b(x)$
- Trigonometric functions: like sine, cosine, and tangent, relate angles to ratios of sides in right triangles.
- Piecewise functions: change their behavior based on specific intervals of input values.
- Absolute value functions: output the positive magnitude of a value regardless of its sign.
- Polynomial functions: involve multiple terms and can have varying degrees.



FURTHER READING:

- "Precalculus" by James Stewart, Lothar Redlin, and Saleem Watson
- "Calculus" by James Stewart
- "Elementary Linear Algebra" by Howard Anton and Chris Rorres
- "Elementary Differential Equations and Boundary Value Problems" by William E. Boyce and Richard C. DiPrima

Online Resources:

MIT OpenCourseWare: Massachusetts Institute of Technology offers open access to various courses, including lectures and materials on functions. [MIT OpenCourseWare - Mathematics](#)



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THANK YOU